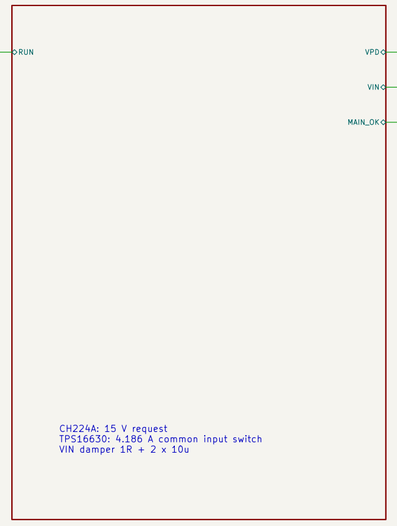
USB POWER S9.0 / SINGLE BOARD, +/−12V ONLY, 15 V PD INPUT

S9.0: +5V subsystem removed; 15 V / 3 A PD input; +12V buck−boost; ceramic banks; two parallel outputs; VPD−powered OV guard (+12V / −12V); one−shot arming latch; MAIN LED only.

USB PD, main eFuse and VIN damper



File: 01_usb_input.kicad_sch

+12 V buck−boost converter



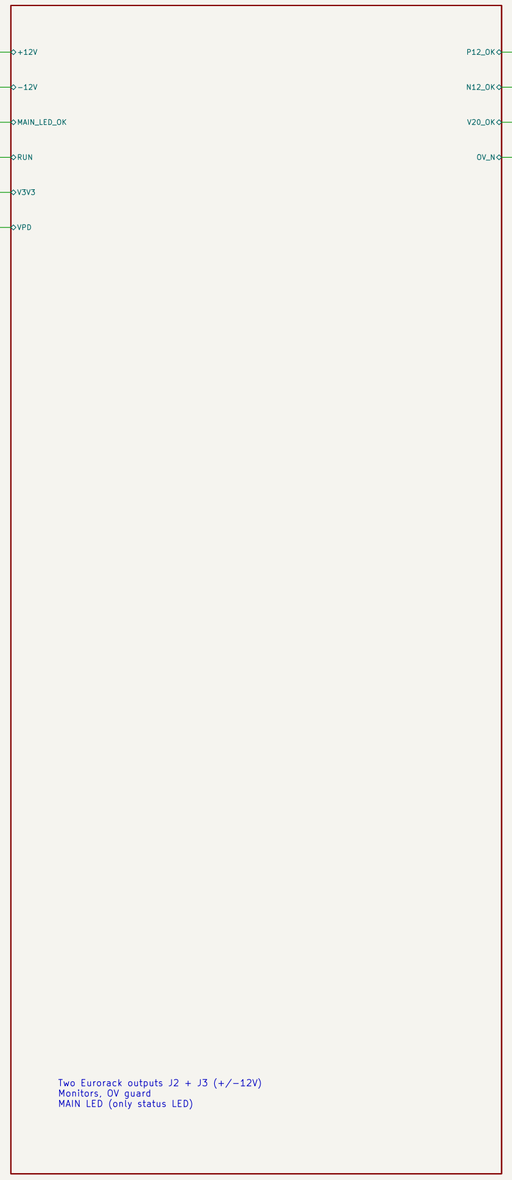
File: 02_positive_12v.kicad_sch

−12 V inverting converter



File: 03_negative_converter.kicad_sch

Two output headers − monitors, overvoltage guard and MAIN LED

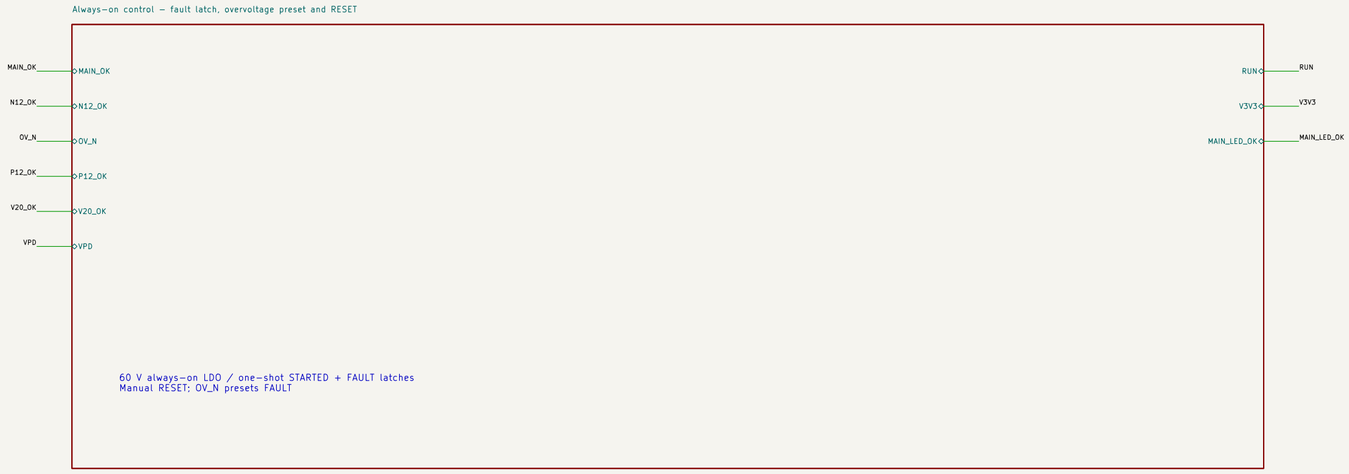


File: 05_outputs_status.kicad_sch

FAULT POLICY
One−shot qualification from the first converter enable (STARTED latch): after
0.31−0.88 s any persistent bad status −> all OFF until RESET. Severe OV via U20 acts even before ARMED.
S9.0: ANY_BAD = NOT(MAIN_LED_OK AND P12_OK AND N12_OK); no +5V rail.
MAIN indicates enabled/status, not precision regulation.

OUTPUTS: +12 / −12 on J2 and J3 in parallel (pins 11/12 open)
Published prototype rating (S9.0): +12V 2.4 A, −12V 0.8 A <= 33 W together,
from a 15 V / 3 A (>= 45 W) USB PD source; above 2 A per rail use both headers.
No reverse blocking: never plug an unpowered module into a live bus.

Always−on control − fault latch, overvoltage preset and RESET



File: 04_fault_reset.kicad_sch

One−shot arming latch (STARTED): U18/U20 on VPD; see docs/s8−single−board.md
S9.0 single board: +/−12V only (+5V subsystem removed); MAIN LED only

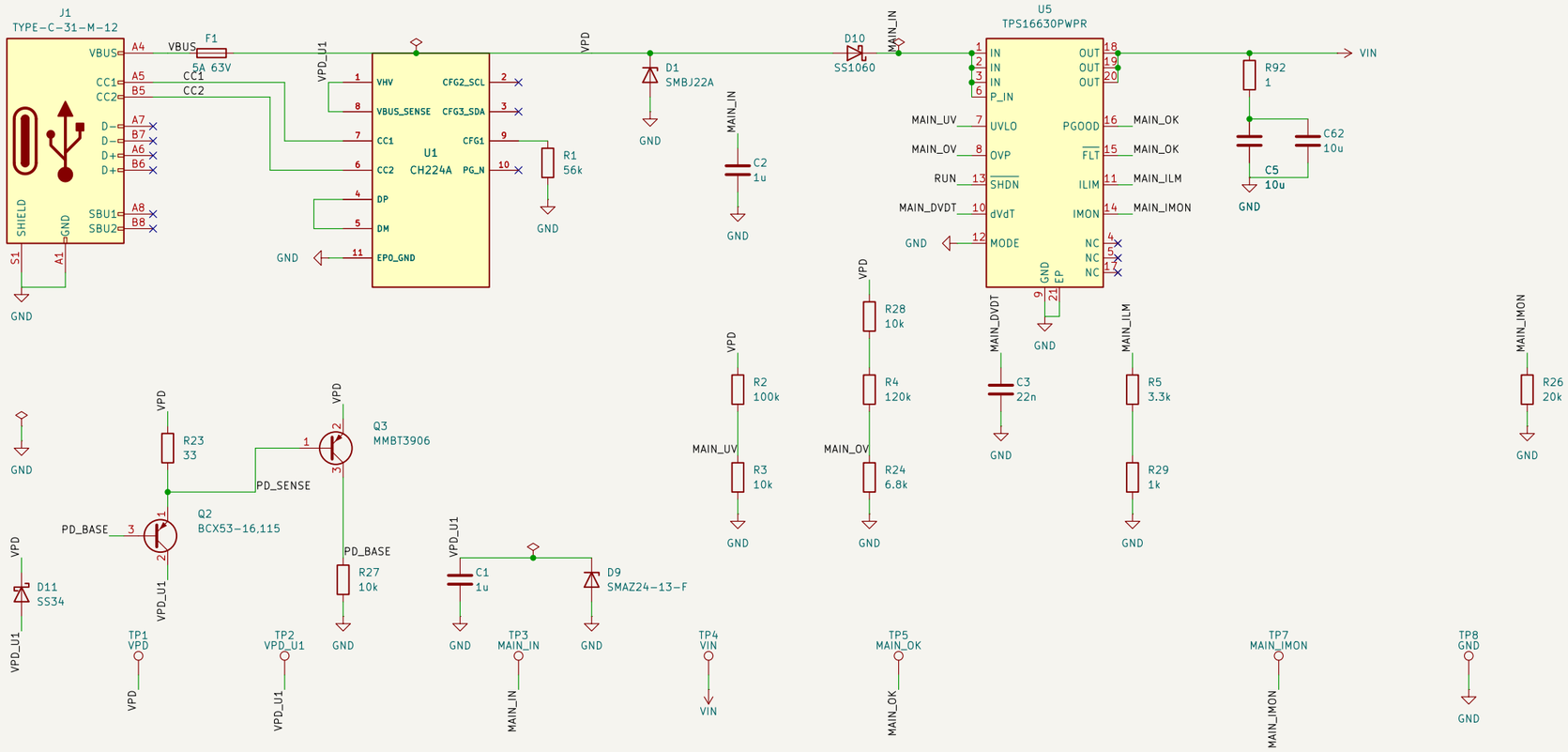
Big Modular

Sheet: /
File: usb−power−45w−single.kicad_sch

Title: USB power S9 single board − wired power flow overview

Size: User	Date:	**Rev: S9.0**
KiCad E.D.A. 10.0.6		Id: 1/6

USB PD, MAIN EFUSE AND VIN DAMPER



S9.0: 15 V PD request (R1 56k; 120k = 20 V); U5 TPS16630 (60 V, 4.186 A nominal) is the common input switch; RUN low turns every rail off. UVLO R2/R3 100k/10k: on 12.46–13.98 V, off 11.55–13.14 V (input qualification). OVP keeps the 20 V–input setting (24.1 V nominal). MAIN_OK = FLT + PGOOD at 3.3 V. R92 1R + C5/C62 2 x 10u damp the VIN bus. The U5 nets are named MAIN_* on the single board (EF_* in S7). Needs a 15 V / 3 A fixed PDO (>= 45 W PD source).

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_OK ◇—— MAIN_OK

RUN ◇—— RUN

VIN ◇—— VIN

VPD ◇—— VPD

S9.0: +/-12V only (+5V removed), one-shot arming latch — docs/s8-single-board.md

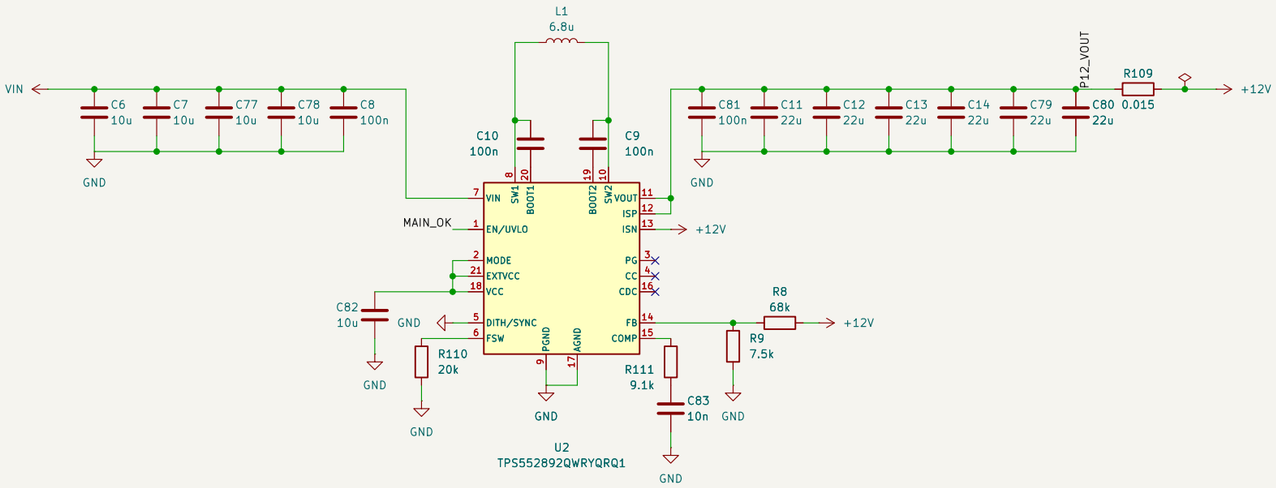
Big Modular

Sheet: /USB PD, main eFuse and VIN damper/
File: 01_usb_input.kicad_sch

Title: USB PD, main eFuse and VIN damper

Size: User	Date:	**Rev: S9.0**
KiCad E.D.A. 10.0.6		Id: 2/6

+12 V BUCK-BOOST CONVERTER





S9.0: TPS552892-Q1 four-switch buck-boost: regulates 12.08 V from the whole converter input (10.55 V U5 sag .. 16.5 V, and 21 V for the later 20 V retarget). The S8 TPS62933F buck needed 0.995-1.053 duty at the 12.5 V low line (1.033 even unloaded), above its 98 % maximum. 966 kHz (R110), L1 6.8 uH, FPWM, internal LDO (MODE/EXTVCC = VCC). R109 15 mR: constant-current limit 3.14-3.53 A = the +12V capability (published 2.4 A): large rack C charges at it. FB R8/R9 after R109: COMP R111/C83 (~10 kHz screen, bench Bode open). C11-C14, C79, C80 + C81 at VOUT; C34 (J2) and C59 (J3) at the headers on page 05. Its VOUT OVP (22.5-24.5 V) is not a 12 V protection (U20 is).

INTER-SHEET INTERFACE (see wired power-flow overview)

+12V ◇ +12V

MAIN_OK ◇ MAIN_OK

VIN ◇ VIN

S9.0: +/-12V only (+5V removed), one-shot arming latch - docs/s8-single-board.md

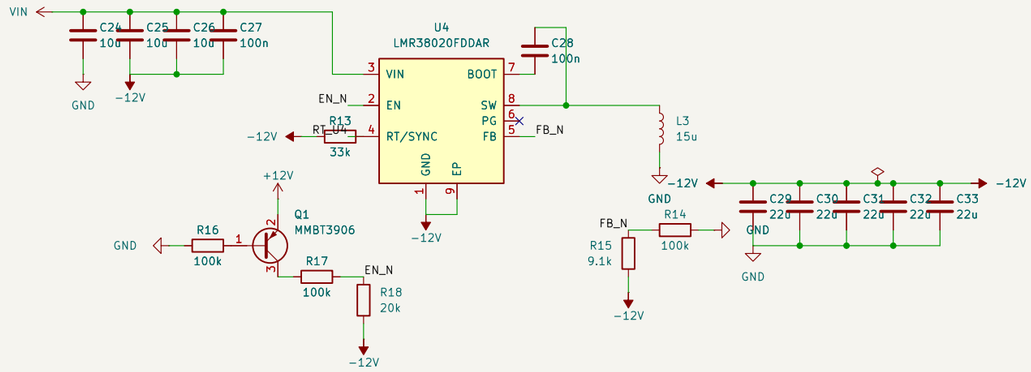
Big Modular

Sheet: /+12 V buck-boost converter/
File: 02_positive_12v.kicad_sch

Title: +12 V buck-boost converter

Size: User	Date:	**Rev: S9.0**
KiCad E.D.A. 10.0.6		Id: 3/6

−12 V INVERTING CONVERTER





LMR38020F inverting buck−boost; IC ground / exposed pad = the delivered −12V (no series switch).
4 ms soft start, 18 ms OCP blanking at start−up. Q1 follows +12V. C29−C33 here + C36 (J2) and C61 (J3).
S8.4: hiccup needs |Vout| < 40 % after the blanking. Rack allowance 0.85 mF (TI 1000 uF COUT guidance; hiccup model
at the U5 turn−on minimum, 0.8 A) covers a typical 0.44 mF rack. S9.0 at 15 V: D 0.44, published 0.8 A.
S9.0 residual: Q1 derives EN_N from the rails only, so a live +/−12V bus can enable U4 in an unpowered module.

INTER−SHEET INTERFACE (see wired power−flow overview)

+12V ◇ +12V

−12V ◇ −12V

VIN ◇ VIN

S9.0: +/−12V only (+5V removed), one−shot arming latch − docs/s8−single−board.md

Big Modular

Sheet: /−12 V inverting converter/
File: 03_negative_converter.kicad_sch

Title: −12 V inverting converter

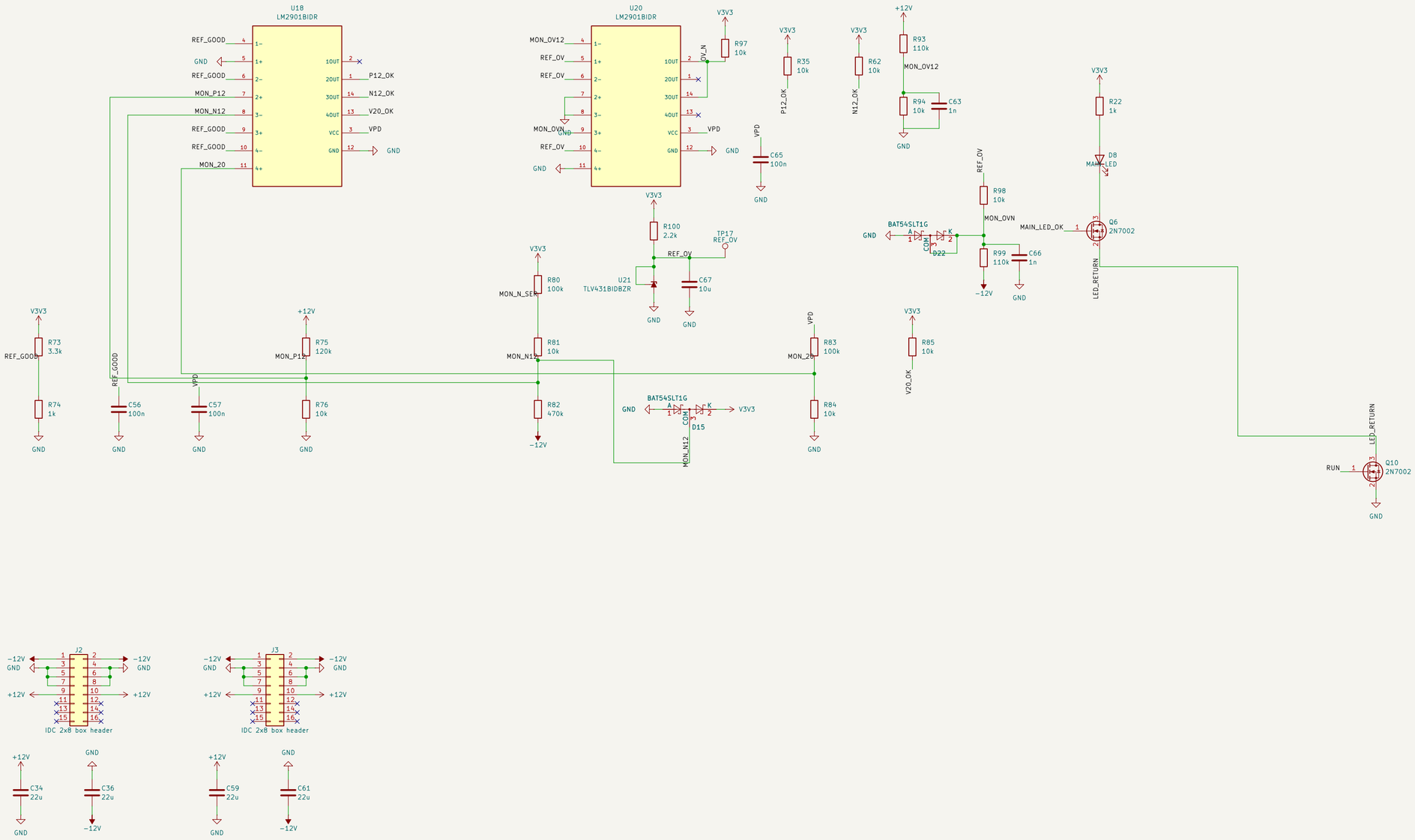
Size: User	Date:	**Rev: S9.0**
KiCad E.D.A. 10.0.6		Id: 4/6

ALWAYS-ON CONTROL — FAULT LATCH, OVERVOLTAGE PRESET AND RESET

U9
TPS7A1633DGNR
C46
1u
C47
10u
R63
10k
MAIN_OK
R64
10k
RUN
U10
TPS3808G01DBVR
R91
15k
POR_SEN
R65
47k
R66
10k
R67
10k
RESET_N
R68
10k
RESET_BTN_N
C48
100n
U11
TPS3808G01DBVR
R69
33k
START_SENSE
TIMER_GO
R70
10k
C49
100n
START_CT
R71
10k
ARMED
C50
100n
SW1
TL3340AF160QG
RESET_BTN_N
R72
10k
ANY_BAD
BAD_RC
C51
100n
U12
SN74HC74DR
STARTED
FAULT_LATCH
NOT_FAULT
FAULT_CLK
C52
100n
U13
SN74HC08DR
TIMER_GO
MAIN_LED_OK
C53
100n
U14
SN74HC20DR
P12_OK
N12_OK
START_N
C54
100n
U15
SN74LVC1G17DBVR
BAD_FILTERED
C55
100n
TP9
RUN
TP10
ARMED
TP11
FAULT_LATCH
TP12
P12_OK
TP14
N12_OK
TP15
V20_OK
TP16
V3V3
TP6
OV_N
TP19
STARTED
RESET clears STARTED and FAULT; RUN is forced low while reset is asserted.
S8.5 one-shot arming: START_N = NAND(MAIN_LED_OK, RUN) (U14 gate 2) presets STARTED (U12 A) at the first converter enable; only RESET clears it. TIMER_GO = RUN AND STARTED -> U11 -> ARMED 0.31–0.88 s later, held until RESET/FAULT: no fault can restart or extend the window. Then any persistent bad status latches every rail OFF. OV_N presets FAULT at any time.
S9.0: ANY_BAD = NAND(MAIN_LED_OK, P12_OK, N12_OK, V3V3): the former +5V input of U14 gate 1 is tied high.
V20_OK (S9.0: RB3 100k, 7.67–9.27 V) only validates U5 FLT/PGOOD (MAIN_OK): the input qualification is the U5 UVLO (on 12.46–13.98 V, off 11.55–13.13 V at the 15 V input).
INTER-SHEET INTERFACE (see wired power-flow overview)
MAIN_LED_OK
MAIN_OK
N12_OK
OV_N
P12_OK
RUN
V20_OK
V3V3
VPD
S9.0: +/−12V only (+5V removed), one-shot arming latch — docs/s8-single-board.md
Big Modular
Sheet: /Always-on control — fault latch, overvoltage preset and RESET/
File: 04_fault_reset.kicad_sch
Title: Always-on control — fault latch, overvoltage preset and RESET
Size: User
Date:
Rev: S9.0
KiCad E.D.A. 10.0.6
Id: 5/6

TWO OUTPUT HEADERS — MONITORS, OVERVOLTAGE GUARD AND MAIN LED

J2 + J3: two parallel 16-pin outputs, pin 1 = −12V. C34/C36 at J2, C59/C61 at J3 (22u each, replace the 100u cans).
S9.0: +/−12V only; J2/J3 pins 11/12 (+5V) and CV/GATE (13–16) are open.
D8 MAIN is the only status LED (rail LEDs removed): MAIN_LED_OK (U5 on, FLT/PGOOD valid) AND RUN. Any bad status after ARMED or a U20 guard trip latches every rail off. P12_OK / N12_OK: logic + test pads.
U20 overvoltage guard (S8.1): +12V > 14.88 V or −12V < −13.64 V nominal (corners 14.48–15.27 / 13.27–14.02 V) pulls OV_N -> FAULT -> U5 off. Own reference REF_OV = U21 TLV431B (R100, C67, TP17); U18/U20 LM2901B: R93, R94, R98, R99 are 0.1 % / 25 ppm. The −12V channel compares MON_OVN with GND (no V3V3 term); D22 holds MON_OVN near −0.24 V (25 C only, <=0.15 mA) if V3V3 is lost while the rack holds −12V. S9.0: the former +5V comparators of U18/U20 are spares.
S8.3: U18/U20 supplies are always-on VPD; outputs remain 3.3 V logic. No prototype qualification is claimed.



INTER-SHEET INTERFACE (see wired power-flow overview)

+12V — +12V
−12V — −12V
MAIN_LED_OK — MAIN_LED_OK
N12_OK — N12_OK
OV_N — OV_N
P12_OK — P12_OK
RUN — RUN
V20_OK — V20_OK
V3V3 — V3V3
VPD — VPD

S9.0: +/−12V only (+5V removed), one-shot arming latch — docs/s8-single-board.md
Big Modular
Sheet: /Two output headers — monitors, overvoltage guard and MAIN LED/
File: 05_outputs_status.kicad_sch
Title: Two output headers — monitors, overvoltage guard and MAIN LED
Size: User | Date: | **Rev: S9.0**
KiCad E.D.A. 10.0.6 | Id: 6/6